

Linear Algebra (MT1004)

Sessional-I Exam

Date: September 24, 2025

Course Instructor(s)

Prof. Abdul Hafeez Sheikh

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Total Time (Hrs): 1 Hour

Total Marks: 50

Total Questions: 1

Roll No

Section

Student Signature

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Instruction/Notes:

1. Programmable calculators are not allowed.
2. Marks will be awarded till the correct calculations, incorrect step voids further marking.
3. Attempt all question parts together. Question attempted in separate parts will not be marked.

CLO #1: Use concept of elementary row operations to find the inverse of square matrices, determinant of a matrix and solving the system of linear equations.

Q#1(a)[10]: Consider the following system in unknowns x , y and z . For what value(s) of "a, b, c" does the following system have no solution? Unique solution? Infinitely many solutions?

$$x + 3y - z = a$$

$$x + y + 2z = b$$

$$2y - 3z = c$$

b)[10]: Use Inversion Algorithm to find the inverse of $A = \begin{bmatrix} 4 & 2 \\ 3 & 5 \end{bmatrix}$ and also write down A and A^{-1} as a product of elementary matrices

$$A = E_1^{-1}E_2^{-1}E_3^{-1} \dots E_{k-1}^{-1}E_k^{-1}, A^{-1} = E_kE_{k-1} \dots E_3E_2E_1.$$

c)[10]: Evaluate the determinant by Elementary Row Operations Method.

$$\begin{vmatrix} 1 & 2 & 3 & 1 \\ -1 & -1 & -1 & 2 \\ 1 & 3 & 1 & 1 \\ -2 & -2 & 0 & -1 \end{vmatrix}$$

d)[10]: Convert the following system into a suitable form and find the number of free variables in the general solution of the linear system also identify the pivot rows and pivot columns. Also, write the general solution in Matrix Column notation.

$$\begin{aligned} x_3 + x_4 + x_5 &= 0 \\ x_1 + x_2 - 2x_3 - x_5 &= 0 \\ 2x_1 + 2x_2 - x_3 + x_5 &= 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \end{aligned}$$

e)[3+3+2+2]: Prove the following statements:

- i) If A and B are invertible matrices with the same size, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
- ii) If A is an invertible matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$.
- iii) Let A and B be symmetric matrices then investigate that whether the product AB is necessarily symmetric. If not give a counter example.
- iv) Prove / disprove that $\det(A + B) = \det(A) + \det(B)$ for any square matrices A and B .

f)[Bonus-5]: State Only the Equivalence Theorem & Its negation.

Good Luck